Complex-valued functions of a real variable are functions $w = f(t)$ where t is a real number and w is a complex number. It is customary to express such functions in terms of two real-valued functions of the real variable t, $w(t) = x(t) + iy(t)$. For example, $w(t) = 3t + i\cos t$ is a complex-valued function of a real variable t.

These special types of complex functions will appear in various places throughout the text.

EXERCISES 2.1

Answers to selected odd-numbered problems begin on page ANS-6.

In Problems 1–8, evaluate the given complex function f at the indicated points.

1. $f(z) = z^2\bar{z} - 2i$	**(a)** $2i$	**(b)** $1 + i$	**(c)** $3 - 2i$
2. $f(z) = -z^3 + 2z + \bar{z}$	**(a)** i	**(b)** $2 - i$	**(c)** $1 + 2i$
3. $f(z) = \log_e \lvert z\rvert + i\,\mathrm{Arg}(z)$	**(a)** 1	**(b)** $4i$	**(c)** $1 + i$
4. $f(z) = \lvert z\rvert^2 - 2\mathrm{Re}(iz) + z$	**(a)** $3 - 4i$	**(b)** $2 - i$	**(c)** $1 + 2i$
5. $f(z) = (xy - x^2) + i(3x + y)$	**(a)** $3i$	**(b)** $4 + i$	**(c)** $3 - 5i$
6. $f(z) = e^z$	**(a)** $2 - \pi i$	**(b)** $\frac{\pi}{3}i$	**(c)** $\log_e 2 - \frac{5\pi}{6}i$
7. $f(z) = r + i\cos^2\theta$	**(a)** 3	**(b)** $-2i$	**(c)** $2 - i$
8. $f(z) = r\sin 3\theta + i\cos 2\theta$	**(a)** -2	**(b)** $1 + i$	**(c)** $-5i$

In Problems 9–16, find the real and imaginary parts u and v of the given complex function f as functions of x and y.

9. $f(z) = 6z - 5 + 9i$

10. $f(z) = -3z + 2\bar{z} - i$

11. $f(z) = z^3 - 2z + 6$

12. $f(z) = z^2 + \bar{z}^2$

13. $f(z) = \dfrac{\bar{z}}{z + 1}$

14. $f(z) = z + \dfrac{1}{z}$

15. $f(z) = e^{2z+i}$

16. $f(z) = e^{z^2}$

In Problems 17–22, find the real and imaginary parts u and v of the given complex function f as functions of r and θ.

17. $f(z) = \bar{z}$

18. $f(z) = \lvert z\rvert$

19. $f(z) = z^4$

20. $f(z) = z + \dfrac{1}{z}$

21. $f(z) = e^z$

22. $f(z) = x^2 + y^2 - yi$

In Problems 23–26, find the natural domain of the given complex function f.

23. $f(z) = 2\mathrm{Re}(z) - iz^2$

24. $f(z) = \dfrac{3z + 2i}{z^3 + 4z^2 + z}$

25. $f(z) = \dfrac{iz}{\lvert z - 1\rvert}$

26. $f(z) = \dfrac{iz}{\lvert z\rvert - 1}$

Focus on Concepts

27. Discuss: Do the following expressions define complex functions $f(z)$? Defend your answer.

(a) $\arg(z)$ **(b)** $\mathrm{Arg}(z)$ **(c)** $\cos(\arg(z)) + i\sin(\arg(z))$

(d) $z^{1/2}$ **(e)** $\lvert z\rvert$ **(f)** $\mathrm{Re}(z)$

28. Find the range of each of the following complex functions.

(**a**) $f(z) = \text{Im}(z)$ defined on the closed disk $|z| \le 2$

(**b**) $f(z) = (1+i)\,|z|$ defined on the square $0 \le \text{Re}(z) \le 1, 0 \le \text{Im}(z) \le 1$

(**c**) $f(z) = \bar{z}$ defined on the upper half-plane $\text{Im}(z) > 0$

29. Find the natural domain and the range of each of the following complex functions.

(**a**) $f(z) = \dfrac{z}{|z|}$. [*Hint*: In order to determine the range, consider $|f(z)|$.]

(**b**) $f(z) = 3 + 4i + \dfrac{5z}{|z|}$.

(**c**) $f(z) = \dfrac{z+\bar{z}}{z-\bar{z}}$.

30. Give an example of a complex function whose natural domain consists of all complex numbers except 0, $1+i$, and $1-i$.

31. Determine the natural domain and range of the complex function $f(z) = \cos(x-y) + i\sin(x-y)$.

32. Suppose that $z = x + iy$. Reread Section 1.1 and determine how to express x and y in terms of z and $\bar{z}$. Then write the following functions in terms of the symbols z and $\bar{z}$.

(**a**) $f(z) = x^2 + y^2$ (**b**) $f(z) = x - 2y + 2 + (6x+y)i$

(**c**) $f(z) = x^2 - y^2 - (5xy)i$ (**d**) $f(z) = 3y^2 + \left(3x^2\right)i$

33. In this problem we examine some properties of the complex exponential function.

(**a**) If $z = x + iy$, then show that $|e^z| = e^x$.

(**b**) Are there any complex numbers z with the property that $e^z = 0$? [*Hint*: Use part (a).]

(**c**) Show that $f(z) = e^z$ is a function that is periodic with pure imaginary period $2\pi i$. That is, show that $e^{z+2\pi i} = e^z$ for all complex numbers z.

34. Use (3) to prove that $e^{\bar{z}} = \overline{e^z}$ for all complex z.

35. What can be said about z if $\left|e^{-z}\right| < 1$?

36. Let $f(z) = \dfrac{e^{iz} + e^{-iz}}{2}$.

(**a**) Show that f is periodic with real period 2π.

(**b**) Suppose that z is real. That is, $z = x + 0i$. Use (3) to rewrite $f(x+0i)$. What well-known real function do you get?

37. What is the period of each of the following complex functions?

(**a**) $f(z) = e^{z+\pi}$ (**b**) $f(z) = e^{\pi z}$

(**c**) $f(z) = e^{2iz}$ (**d**) $f(z) = e^{3z+i}$

38. If $f(z)$ is a complex function with pure imaginary period i, then what is the period of the function $g(z) = f(iz-2)$?

Computer Lab Assignments

In Problems 39–42, use a CAS to evaluate the given complex function at the indicated points. In *Mathematica* you can define a function using the syntax **f[z_]:=** *an expression in the variable* z. To evaluate the function at the point $a+bi$, use the syntax **f[a+b*I]**.

39. the function and points in Problem 1.

40. the function and points in Problem 2.

41. the function and points in Problem 3.

42. the function and points in Problem 4.

2.2 Complex Functions as Mappings

INTRODUCTION Recall that if f is a real-valued function of a real variable, then the graph of f is a curve in the Cartesian plane. Graphs are used extensively to investigate properties of *real* functions in elementary courses. However, we'll see that the graph of a *complex* function lies in four-dimensional space, and so we cannot use graphs to study complex functions. In this section we discuss the concept of a *complex mapping*, which was developed by the German mathematician Bernhard Riemann to give a geometric representation of a complex function. The basic idea is this. Every complex function describes a correspondence between points in two copies of the complex plane. Specifically, the point z in the z-plane is associated with the unique point $w = f(z)$ in the w-plane. We use the alternative term **complex mapping** in place of "complex function" when considering the function as this correspondence between points in the z-plane and points in the w-plane. The geometric representation of a complex mapping $w = f(z)$ consists of two figures: the first, a subset S of points in the z-plane, and the second, the set S' of the images of points in S under $w = f(z)$ in the w-plane.

Mappings

A useful tool for the study of real functions in elementary calculus is the graph of the function. Recall that if $y = f(x)$ is a real-valued function of a real variable x, then the graph of f is defined to be the set of all points $(x, f(x))$ in the two-dimensional Cartesian plane. An analogous definition can be made for a complex function. However, if $w = f(z)$ is a complex function, then both z and w lie in a complex plane. It follows that the set of all points $(z, f(z))$ lies in four-dimensional space (two dimensions from the input z and two dimensions from the output w). Of course, a subset of four-dimensional space cannot be easily illustrated. Therefore:

- *We cannot draw the graph of a complex function.*

The concept of a complex mapping provides an alternative way of giving a geometric representation of a complex function. As described in the section introduction, we use the term **complex mapping** to refer to the correspondence determined by a complex function $w = f(z)$ between points in a z-plane and images in a w-plane. If the point z_0 in the z-plane corresponds to the point w_0 in the w-plane, that is, if $w_0 = f(z_0)$, then we say that f **maps** z_0 onto w_0 or, equivalently, that z_0 is **mapped** onto w_0 by f.

As an example of this type of geometric thinking, consider the real function $f(x) = x + 2$. Rather than representing this function with a line of slope 1 and y-intercept (0, 2), consider how one copy of the real line (the x-line) is mapped onto another copy of the real line (the y-line) by f. Each point on the x-line is mapped onto a point two units to the right on the y-line (0 is mapped onto 2, while 3 is mapped onto 5, and so on). Therefore, the real function $f(x) = x + 2$ can be thought of as a mapping that *translates* each point in the real line two units to the right. You can visualize the action of this mapping by imagining the real line as an infinite rigid rod that is physically moved two units to the right.

In order to create a geometric representation of a complex mapping, we begin with two copies of the complex plane, the z-plane and the w-plane, drawn either side-by-side or one above the other. A complex mapping is represented by drawing a set S of points in the z-plane and the corresponding set of images of the points in S under f in the w-plane. This idea is illustrated in **FIGURE 2.2.1** where a set S in the z-plane is shown in color in Figure 2.2.1(a) and a set labeled S', which represents the set of the images of points in S under $w = f(z)$, is shown in gray in Figure 2.2.1(b). From this point on we will use notation similar to that in Figure 2.2.1 when discussing mappings.